

ADA – PLC Guide

As-Built Notes on the Ladder Logic

Nebico Oil Melting/Retention System – Version 2.0 (detailed edition)

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This edition builds on the V1.1 guide and walks through Program 2.1, which adds proper sensor-reading logic ahead of everything V1.1 already covered, fixes the solar-gate comparison, and – for the first time – puts a real name on every filtered sensor register. A few loose ends turned up while working through the new sensor-reading chain; most have now been checked and closed off, and the couple that remain are noted at the end.

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1 Purpose and Scope

This guide builds on *ADA – PLC Guide, Version 1.1* and covers **PLC Program 2.1**, exported 9 August 2026. The programme’s own revision notes mention three things: sensor-reading logic is now included ahead of the process logic V1.1 already covered; the correlation between the HMI and the PLC is fixed; and the solar-gate condition is fixed. On top of that, this edition also puts a proper name on every filtered sensor register (D120–D138 and the D18–D36 raw registers behind them), which V1.1 could only call “filtered sensor values.”

- Section 2 sets out the sensor identity map that the rest of this document relies on.
- Section 4 goes through the programme as it stands now, start to finish. Green boxes mark anything new or newly understood since V1.1.
- Section 7 looks at what actually changed since V1.1 – the new sensor front end, the solar-gate fix, and a couple of things worth a second look, now fixed.
- Section 8 carries the running list of placeholder values from V1.1 and adds the new sensor-side ones.

2 Sensor Channel Identity Map

V1.1 could only call D120–D138 “ten channels, filtered,” with no way to say what any of them actually measured. This edition puts a real name on each one. The table below is used right through the rest of the document, and it also corrects two spots where the V1.1 write-up named the wrong register for something the ladder was already reading correctly.

Raw reg.	Filtered reg.	Quantity	Averaging block
D18	D120	Temp T5	Sensor Average 1 (PLC, ch. AD0)
D20	D122	Temp TA	Sensor Average 1 (PLC, ch. AD1)
D22	D124	Temp TB	Sensor Average 1 (4RTD-1, ch. 1)
D24	D126	Temp TC	Sensor Average 1 (4RTD-1, ch. 2)
D26	D128	Temp HEA	Sensor Average 1 (4RTD-1, ch. 3)
D28	D130	Temp SOL	Sensor Average 2 (4RTD-1, ch. 4)
D30	D132	Level T5	Sensor Average 2 (4AD-2, ch. 1)
D32	D134	Level TA	Sensor Average 2 (4AD-2, ch. 2)
D34	D136	Level TB	Sensor Average 2 (4AD-2, ch. 3)
D36	D138	Level TC	Sensor Average 2 (4AD-2, ch. 4)

Corrects V1.1. The V1.1 write-up described the Water Temperature Target Core Logic as reading “level D120” and Solar Gating as comparing against “D136, filtered solar temperature.” The ladder itself has always used D132 and D128 – that part of the code never changed. With the table above, D132 turns out to be Level T5 (which makes far more sense for a level-driven target-temperature rule) and D128 is Temp HEA. V1.1 was written before this mapping existed, so it made a reasonable guess at the names. This edition swaps in the confirmed names and puts both sections right.

Still open. D282 is the empty-tank level reference for the T5 ultrasonic channel; D284 is the shared empty-tank reference for TA/TB/TC, since all three tanks are the same size. Both only appear in the new Ultrasonic Math block (§4) and neither has a proper commissioned value yet – see §8.

3 Program Structure Overview

The programme's own step index, as it stands in V2.1. The bold rows are new in V2.1 – the sensor-reading front end. Everything else is the same set of named blocks V1.1 already covered, just renumbered because roughly 400 extra steps of sensor-reading logic now sit ahead of them.

Step	Section
0	RTD From PLC
26	RTD From Slot 1
56	Temperature Math
213	AD from Slot 2
243	Ultrasonic Math
400	System Install
423	Timer for System
437	MODBUS Logic
640	Automation
643	Physical E-Switch
646	Virtual E-Switches
662	System OK
667	Sensor Average 1
862	Sensor Average 2
1055	Water Temperature Target Core Logic
1167	Solar Gating
1193	Heater Latching
1395	Height-to-Mass Conversion
1448	Dynamic Upper Thermal Boundary (T5)
1524	PO Designation
1594	PO Melting Timer
1776	PO Scores
1894	Priority Scheduler
1966	Mode Scheduler
2006	Mode Display in HMI
2030	PO Designated?
2034	PO Score Comparison
2130	Need Array Core
2496	Oil Distribution
2820	Sensor Change Rate Calculations
3051	Fault Analysis (High)
3162	Fault Analysis (Calculations for Low)
3293	Fault Analysis (Low)
3364	Oil Distribution Chart
3438	Jacket Pump
3440	AODD Pump
3513	Manual Oil Pump
3521	HMI Display for Inputs
3585	HMI Charting
3622	Mode Adjuster for Manual Control
3712	END

4 Functional Walkthrough

4.1 RTD From PLC (Step 0)

On the always-on bit M8000, the two onboard analogue channel words are copied out of their special registers into working memory (D8030 → D222, D8031 → D224) and converted to floating point (FLT) into D226 and D228. This is where the PLC’s own built-in analogue channels are read – the source that sat behind what V1.1 simply called “raw sensor inputs.”

New in V2.1. Did not exist in V1.1. The raw inputs V1.1’s register map listed at D17–D26 (now D18–D28, see §2) used to come from somewhere unspecified. This block, together with RTD From Slot 1, Temperature Math, AD from Slot 2, and Ultrasonic Math below, is that source, spelled out properly for the first time.

4.2 RTD From Slot 1 (Step 26)

FROM K1 K2 D230 K4 reads four words from the buffer memory of the module in slot 1 into D230–D233 – a second bank of raw temperature channels, alongside the two read directly above.

4.3 Temperature Math (Step 56)

D230–D233 are converted to floating point (D234, D236, D238, D240). All six raw channels – the two onboard ones plus these four – are then scaled by DEMUL ... E0.1 and passed through DESUB ... E1.1223 to land in D18, D20, D22, D24, D26, D28 – the raw temperature values feeding Sensor Average 1 and the first channel of Sensor Average 2 (Temp SOL). E0.1 fits a tenths-of-a-degree reading; E1.1223 is a fixed offset applied the same way across all six channels.

Fixed. An earlier pass through this chain found that the scaling for Temp TC was landing in D148 instead of D248, leaving D24 (and Temp TC downstream) reading a stale, never-updated value. That has now been corrected in the programme, so this description matches what the ladder actually does: all six channels scale and land in their own register, D18 through D28, exactly as the table above shows. No separate write-up of the old fault is needed here any more.

4.4 AD from Slot 2 (Step 213)

FROM K2 K2 D254 K4 reads four words from the module in slot 2 into D254–D257 – the raw ultrasonic level channels for T5, TA, TB, and TC.

4.5 Ultrasonic Math (Step 243)

D254–D257 are converted to floating point into D258–D264, scaled by DEMUL ... E-0.227 into D266–D272, and offset by DEADD ... E688.884 into D274–D280 – a straightforward linear conversion from raw ultrasonic counts to a distance. That distance is then taken away from an empty-tank reference to turn *distance* into *level*: channel 1 (T5) uses DESUB D282 D274 D30, while channels 2–4 (TA, TB, TC) use the shared reference DESUB D284 ... into D32, D34, D36.

Matches the site mapping. D282 (T5’s own empty-tank reference) and D284 (the shared TA/TB/TC reference, since the three tanks are the same size) line up exactly with what was described on site – and this is also why T5 gets its own reference register while TA/TB/TC share one between them.

4.6 System Install (Step 400)

On the first-scan pulse (M8002), the programme writes fixed values into D8397, D8400, D8401 and sets M8192 – routine set-up of the PLC’s built-in communication port. MOV K0 D550 sets the Modbus write-back register used in §4.20 to zero. Otherwise unchanged from V1.1, just moved further down the step count.

4.7 Timer for System (Step 423)

Unchanged from V1.1: T0 (K1) gates T1 (K5, self-resetting – the RST T1 at the end of Sensor Average 1 confirms this), which gates T2 (K60, self-resetting). T1 stays the master “logic enabled” tick that nearly every later section relies on.

4.8 MODBUS Logic (Step 437)

Unchanged from V1.1: the three-station round-robin poll on ADPRW, gated by T10–T61 and stepped by M0–M5, with D0/D2/D4 unpacked into M51–M59.

4.9 Automation, Physical E-Switch, Virtual E-Switches, System OK (Steps 640–662)

Unchanged from V1.1: X001/X000 mirror into M61/M60; ten fault bits (M83–M92) OR into M62; M63 (SYSTEM_OK) comes from M60, M61, M62, and T1 together.

4.10 Sensor Average 1 & 2 (Steps 667–1054)

Same structure as V1.1: two five-sample rolling averages, block 1 (D18–D26 → D100–D108 → D120–D128, counter D60) and block 2 (D28–D36 → D110–D118 → D130–D138, counter D70).

New in V2.0 – settles an open point. V1.1 could only say “ten channels, filtered.” Per §2, block 1 is Temp T5/TA/TB/TC/HEA and block 2 is Temp SOL plus Level T5/TA/TB/TC. Every block further down that reads D120–D138 (Water Temperature Target Core, Solar Gating, Heater Latching, Height-to- Mass Conversion, Dynamic Upper Thermal Boundary) is described below with this in mind.

4.11 Water Temperature Target Core Logic (Step 1055)

T_{target} (D150) against **Level T5 (D132)**: flat at 70.0°C below $d = 200$, flat at 40.0°C above $d = 360$, with a curve between the two.

Corrects V1.1’s write-up (see §2). The ladder has always compared against D132, not D120 as V1.1’s walkthrough said. Now that D132 is confirmed as Level T5, this block is exactly what its name suggests: a level-driven target-temperature rule, not a temperature-driven one.

Calibration item, carried from V1.1. K200 and K360 stand for 50% and 90% of T5’s usable level range. Still needs a real, site-measured value. See Section 8.

4.12 Solar Gating (Step 1167)

D150 (T_{target}) against **Temp HEA (D128)** routes to Y020 (high side) or Y021 (low side), both still gated by M63 (SYSTEM_OK).

Corrects V1.1’s write-up; likely what the solar-gate fix was about. V1.1’s walk-through named this comparison as running against “D136, filtered solar temperature.” The ladder compares D150 against D128 (Temp HEA), and this looks unchanged across V1.0/V1.1/V2.1 – only the write-up was wrong. The programme’s own note about a fixed solar-gate condition may well be describing this same correction being made in an earlier working copy, though there is no V1.0 ladder export on hand to check that against directly.

4.13 Heater Latching (Step 1193)

T5/T6 sample **Temp T5 (D120)** into shadow register D140 and compare the current-to-previous ratio via DEDIV D120 D140 D148 against E1 for a stability flag (M210/M215/M220), combined with banded compares of **Level T5 (D132)** against K55/K80/K95 (rising) and K85/K70/K45 (falling) to give the 0–3 heater-stage count (D152) driving Y010/Y011/Y012.

Corrects V1.1’s generic description. V1.1 only said this block sampled “D120” and banded “D132,” without saying what either one was. With the identity map, this reads as a sensible design: a temperature-stability check (Temp T5, ratio-based, filters out sensor noise) combined with a level-based staging rule (Level T5, percentage bands).

4.14 Height-to-Mass Conversion (Step 1395)

Four level values – **Level T5 (D132)**, **Level TA (D134)**, **Level TB (D136)**, **Level TC (D138)** – scaled by DEMUL ... K1 into mass proxies D160–D166.

Calibration item, carried from V1.1. Gain is still unity (K1) on all four channels – a placeholder, not a proper height-to-mass conversion yet. See Section 8.

4.15 Dynamic Upper Thermal Boundary / T5 (Step 1448)

T_f (D178) via the mass-weighted mixing formula, using constant E239.53, combining **Temp SOL (D130)** and the T5 mass proxy (D160) against **Temp T5 (D120)**.

$239.53 = 479060/2000$, confirmed exactly against *ADA Architecture II* §5, unchanged from V1.1.

4.16 PO Designation (Step 1524)

M12/M13/M14 select a tank’s mass and temperature into D168/D180: TA uses mass proxy D162 and **Temp TA (D122)**; TB uses D164 and **Temp TB (D124)**; TC uses D166 and **Temp TC (D126)**. With the Temperature Math fix noted above, all three tanks now read a live, updating temperature here.

4.17 PO Melting Timer (Step 1594)

A two-phase demand function (melting/liquid) into D192, same structure as V1.1. M20 sets when $D192 > 0$. The watchdog (T130, gated on T1 AND D192) still clears D192, D16, and M15 on timeout; M17 (main HMI reset) still clears D16 in parallel.

Fixed. The demand equation used to subtract the selected tank's working temperature (D180) from a fixed register, D138 – Level TC – regardless of which tank (TA/TB/TC) was actually selected. That has now been corrected so the equation uses the selected tank's own level, not a fixed one. The rest of the block – the watchdog, the reset paths – is unchanged from V1.1.

Calibration item, carried from V1.1. E135 and E30 (melting- point / target comparison constants) should be checked against the actual palm-oil product specification. See Section 8.

4.18 PO Scores, Priority Scheduler, Mode Scheduler, Mode Display in HMI (Steps 1776–2029)

Unchanged from V1.1: weighted tank scores (0.6 level / 0.4 temperature) into D212/D214/D216; priority flags M250/M251/M252 (with the V1.1 pause-gate term on M251 via M16); Y002/Y003/Y004 latched by M280/M285/M290; mode-name string written to D480.

4.19 PO Designated?, PO Score Comparison, Need Array Core (Steps 2030–2495)

Unchanged from V1.1: the combined designation flag, the running highest/second-highest score extraction into D218/D220, and the twelve-flag Need Array block driving Y011–Y014.

4.20 Oil Distribution (Step 2496)

Unchanged from V1.1: the FIFO-based (SFWR/SFRD) destination request queue, the DEXP-based dispense-time estimate, the D550 write-back exchange (M330/M331/M332, T12/T13/T14), and the M50/M52/M55/M58 oil-pump interlock with M53/M56/M59 cancel handling.

Likely what the HMI–PLC correlation note refers to. Programme 2.1's own notes say the correlation between the HMI and the PLC is fixed. The D550 write-back exchange here – previously noted as having a payload whose exact meaning wasn't obvious from the ladder alone – is unchanged in this export, so the fix most likely sits on the HMI/GOT side rather than in this block. Worth a quick confirmation so this section can record it properly next time.

4.21 Sensor Change Rate Calculations, Fault Analysis High/Low (Steps 2820–3363)

Unchanged from V1.1: roughly 5s snapshot into D400–D418; ratio checks against D500–D514 (channels 1–4, adjustable) and fixed K2/K1 (channels 5–10) feeding fault bits M83–M92.

4.22 Oil Distribution Chart, Jacket Pump, AODD Pump, Manual Oil Pump (Steps 3364–3520)

Unchanged from V1.1: 24-hour rollover on M8014; jacket pump on M63 without condition; AODD pump tied to PO designation plus the K30 threshold; manual pushbutton override with the same M52/M55/M58/M50 interlock added in V1.1.

4.23 HMI Display for Inputs, HMI Charting, Mode Adjuster for Manual Control (Steps 3521–3711)

Unchanged from V1.1: the 32-point input mirror at M115–M146; the four-value trend-chart block (D6–D9); the manual mode-adjuster path (M8–M11, M610) alongside the automatic Need Array logic.

5 Timer/Counter Presets

Device	Preset	Role
T0	K1	Power-up settling delay
T1	K5	Master logic-enabled tick (self-resetting)
T2	K60	Self-resetting housekeeping pulse
T10, T20, T30, T40, T50, T60	K1/K2	MODBUS sequencer, stations 1–3
T11, T21, T31, T41, T51, T61	K1/K2	MODBUS sequencer, secondary exchange
T5	K3000	Heater-latch sample timer
T6	K3030	Heater-latch sample timer
T100	D330 (calculated)	Oil dispense duration – register preset
T110	K50	Sensor change-rate sample tick (~ 5 s)
T12	K30	Write-back exchange timeout, station 1
T13	K30	Write-back exchange timeout, station 2
T14	K30	Write-back exchange timeout, station 3
T130	TBD	PO-melt-demand reset timer – preset value still not in this export

6 Reconstructed Register Map

Table 1: Reconstructed Register Map – Data Registers (*D*)

Register Range	Inferred Role
D6–D9	HMI trend-chart mirror
D16	Priority-scheduler condition register, cleared by T130/M17
D18, D20, D22, D24, D26	(V2.0) Raw Temp T5/TA/TB/TC/HEA (D24 – see §7)
D28	(V2.0) Raw Temp SOL
D30, D32, D34, D36	(V2.0) Raw Level T5/TA/TB/TC
D60, D70	Sample counters, sensor averaging
D100–D118	Rolling-sum accumulators
D120	(V2.0) Filtered Temp T5
D122	(V2.0) Filtered Temp TA
D124	(V2.0) Filtered Temp TB
D126	(V2.0) Filtered Temp TC (see §7)
D128	(V2.0) Filtered Temp HEA
D130	(V2.0) Filtered Temp SOL
D132	(V2.0) Filtered Level T5
D134	(V2.0) Filtered Level TA
D136	(V2.0) Filtered Level TB
D138	(V2.0) Filtered Level TC
D140, D148	Heater-latch stability sample/ratio
D150	T_{target}
D152	Heater stage count, 0–3
D160–D166	Mass proxies (unity gain – calibration item)
D168, D180	Selected PO tank working temperature/mass
D170–D178	T_f calculation
D182–D192	PO melting/liquid demand function
D200–D216	Weighted tank scores
D218, D220	Running highest/second score
D222, D224	(V2.0) Onboard analog channel raw copies
D226, D228	(V2.0) Onboard analog channels, float
D230–D233	(V2.0) Slot-1 module raw channels
D234–D240	(V2.0) Slot-1 module channels, float
D242–D252	(V2.0) Temperature-math scratch (scale/offset chain)
D254–D257	(V2.0) Slot-2 module raw channels (ultrasonic)
D258–D264	(V2.0) Ultrasonic channels, float
D266–D280	(V2.0) Ultrasonic scale/offset scratch
D282	(V2.0) Empty-tank level reference, T5
D284	(V2.0) Empty-tank level reference, TA/TB/TC (shared)
D300	Oil destination FIFO
D305	Dequeued active destination
D310–D330	Dispense quantity / runtime-estimate chain
D400–D418	Previous-value shadow registers
D420–D438	Rate-of-change ratios
D440–D458	Low-side ratio checks
D460–D462	Cumulative oil consumption
D470	Running minute counter (24 h rollover)
D480	HMI mode-name string display
D500, D502, D504, D506	Adjustable fault high-thresholds, channels 1–4
D508, D510, D512, D514	Adjustable fault low-multipliers, channels 1–4
D550	Modbus write-back value to active satellite

Table 2: Reconstructed Register Map – Internal Relays (*M*)

Register Range	Inferred Role
M0–M5	MODBUS sequencer state
M8–M11	Manual mode-adjuster inputs
M12–M14	PO tank select
M16	Pause button, main HMI
M17	Reset button, main HMI
M50	Master oil switch, master HMI
M51–M59	Unpacked satellite mirror bits
M60–M63	Physical/Virtual E-Switch, System OK
M83–M92	Fault bits, 10 channels
M100–M114	Heater/priority/scheduler intermediate flags
M115–M146	Physical input mirror, X000–X037
M200–M230	Heater hysteresis and stability latches
M250–M300	Priority/mode scheduler latches
M270–M272	PO designation latches
M323, M324	Oil FIFO dequeue handshake
M330, M331, M332	Write-back exchange latches
M470	Daily total reset pulse
M610	Mode-adjuster gating flag

7 Changes Made Since V1.1

7.1 A Proper Sensor Front End

V1.1’s register map listed D17–D26 (now D18–D28 in this edition) as “raw sensor inputs” without saying where they came from. Programme 2.1 answers that: five new named blocks – RTD From PLC, RTD From Slot 1, Temperature Math, AD from Slot 2, and Ultrasonic Math (§4) – now read two onboard analogue channels and two special-function modules (slot 1 for RTD/temperature, slot 2 for ultrasonic level), scale and linearise them, and pass on exactly the raw registers Sensor Average 1/2 were already expecting. Nothing further down the programme changed to make room for this; it was built to feed the existing process logic without touching it.

7.2 Register Identities, Not Just Register Numbers

The main addition in this edition is not really a code change at all – it is the confirmed mapping in §2, tying D120–D138 to real channel names for the first time. That let two V1.1 write-up errors get corrected (Water Temperature Target Core and Solar Gating, both in §4), and it also brought to light two register-usage issues that were impossible to spot without the mapping: the dead register behind Temp TC, and the fixed D138 term in the PO Melting Timer’s demand equation. Both have since been fixed in the programme and are noted as such in §4.

7.3 Solar Gate Fix

The programme’s own notes say the solar-gate condition is fixed. The ladder now compares D150 against D128 (Temp HEA), matching what the logic already needed to be doing – V1.1’s write-up had simply described it as D136. There is no earlier ladder export on hand to check the exact change against, so this document records the current (fixed) state with confidence, but not the precise nature of the fix itself.

7.4 What Stayed the Same

Every block from System Install (Step 400) onward that V1.1 already covered – MODBUS Logic, the E-Switch chain, PO Designation and Scoring, the Priority/Mode Schedulers, Need

Array Core, Oil Distribution (including the V1.1 write-back exchange and pump interlocks), Fault Analysis, and the three HMI sections added in V1.1 – is structurally the same in this export. The only difference is the step numbers, which move down by 400–700 steps because of the new sensor front end and its working registers ahead of them.

8 Calibration Items

Everything below is a constant sitting in the ladder that stands in for a real, physical, site value. Items carried over from V1.1 are unchanged; new ones are bolded.

Location	Constant(s)	What it needs to become
Ultrasonic Math	E-0.227 (scale), E688.884 (offset)	Raw-count-to-distance conversion for the ultrasonic modules – check against the sensor’s own datasheet scaling.
Ultrasonic Math – empty-tank references	D282 (T5), D284 (TA/TB/TC, shared)	Registers rather than constants, but neither has a proper commissioned value yet – the empty-tank distance measurement is still owed.
Water Temperature Target Core	K200, K360	50% and 90% of T5’s usable level range. Still needs a real, site-measured value at commissioning.
Height-to-Mass Conversion	K1 (DEMUL gain, ×4 channels)	Currently unity gain – needs the real height-to-mass conversion figure per tank, from tank geometry and product density.
PO Melting Timer	E135, E30	Melting-point / target-temperature reference constants – check against the actual palm-oil product specification.
Heater Latching	K55, K80, K95, K85, K70, K45 (banded thresholds on Level T5, D132)	Heater-stage transition points – confirm these match the intended staging bands for T5.
Heater Latching – stability gate	T5/T6 (K3000/K3030), E1 (DECMPI148 threshold)	Sampling window and stability-ratio threshold on Temp T5 – may need adjusting once the real sensor noise is known.
Fault Analysis, channels 5–10	K2 (high), K1 (low multiplier)	Still hardcoded – decide whether these need registering like channels 1–4, or are fine left fixed.
AODD Pump	K30 (level/temperature compare)	Trigger threshold for AODD pump engagement – confirm against real tank levels and temperatures.
Oil write-back exchange	T12/T13/T14 (K30 timeout)	Confirm this timeout window suits the actual Modbus round-trip time to the satellites.
PO Melting demand	D190 (calibration multiplier, already a register)	Already tunable without a programme change – the value to load at commissioning is still open.

Not calibration items – structural constants, left as-is. K5 (sample count), K50 (sensor change-rate tick interval), E0.1 (RTD scale – structural, not a stand-in for a measured quantity), E239.53 (= 479060/2000, confirmed exactly against Architecture II §5), and the MODBUS/timer chains (T0/T1/T2, T10–T61) are timing or structural choices, not stand-ins for a physical measurement, and stay out of the table above.

Checked and confirmed: E1.1223. The RTD offset used across all six temperature channels in Temperature Math (§4) was queried in the previous edition – was one shared offset accurate enough, or did each channel need its own? A single offset has now been confirmed as accurate enough, so no per-channel version is needed.

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